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Open Geospatial Consortium

OGC EMISSION EVENT MODELING LANGUAGE (EMISSIONML) PART I: DATA MODEL

DRAFT STANDARD

DRAFT

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I KEYWORDS

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, emissions, climate change

PREFACE

i. Abstract

<Insert Abstract Text here>

ii. Keywords

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, <tags separated by commas>

iii. Preface

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iv. Security Considerations

No security considerations have been made for this Standard.

v. Submitting organizations

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

Organization name(s)

vi. Submitters

All questions regarding this submission should be directed to the editor or the submitters:

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III SECURITY CONSIDERATIONS

No security considerations have been made for this document.

IV SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- SensorUp Inc.
- Natural Resources Canada
- University of Calgary

1

SCOPE

1 SCOPE

NOTE: Insert Scope text here. Give the subject of the document and the aspects of that scope covered by the document.

2

NORMATIVE REFERENCES

2 NORMATIVE REFERENCES

There are no normative references in this document.

The following normative documents contain provisions that, through reference in this text, constitute provisions of this document. For dated references, subsequent amendments to, or revisions of, any of these publications do not apply. For undated references, the latest edition of the normative document referred to applies.

NOTE: Insert References here. If there are no references, state "There are no normative references".

References are to follow the Springer LNCS style, with the exception that optional information may be appended to references: DOIs are added after the date and web resource references may include an access date at the end of the reference in parentheses. See examples from Springer and OGC below.

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3

TERMS AND DEFINITIONS

3 TERMS AND DEFINITIONS

No terms and definitions are listed in this document.

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word "shall" (not "must") is the verb form used to indicate a requirement to be strictly followed to conform to this Standard and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the 'ModSpec'. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

Emission Event	An Emission Event represents the release of a substance into the atmosphere over a non-zero duration. It is characterized by its start time and end time, the Source Feature from which the substance originated, and the identity and quantity of the emitted substance.

Note: Conceptually, an Emission Event corresponds to an occurrent in upper-ontology terms, as it unfolds over time and has temporal parts. Note: For reporting or accounting purposes, a long-running emission event is often segmented into multiple time-bounded emission events (e.g., annual records). This segmentation reflects reporting conventions and does not imply physical cessation or re-initiation of the emission.

Feature	Abstraction of real world phenomena (Source: ISO 19101-1:2014)
Measurement	Set of operations having the object of determining the value of a quantity (Source: ISO/OGC 19156:2023)
Mechanism	A Mechanism is a classification of the specific physical, chemical, or operational process that constitutes the direct cause or pathway for an emission. It is defined independently of the Source—including its identity, structure, or purpose—allowing Mechanisms to be consistently reused across different types of Source.
Observation	Act carried out by an observer to determine the value of an observable property of an object (feature-of-interest) by using a procedure, with the value provided as the result (Source: ISO/OGC 19156:2023)
Source	A Source represents the feature from which an emission originates. A Source is an ISO 191xx feature whose geometry provides the geospatial reference for one or more Emission Events. A Source is characterised by its own identity, geometry, intended function, and lifecycle, and is explicitly agnostic to the Mechanism by which any associated Emission Event occurs.

Note: A Source is a continuant that may exist independently of any emission, whereas an Emission Event is an occurrent. The identity and lifecycle of a Source are independent of the existence, absence, or frequency of Emission Events.

Unit of Measure	Reference quantity chosen from a unit equivalence group (Source: ISO/OGC 19156:2023)
term name	text of the definition

4

CLAUSES NOT CONTAINING NORMATIVE MATERIAL

4 CLAUSES NOT CONTAINING NORMATIVE MATERIAL

Paragraph

4.1. Clauses not containing normative material sub-clause 1

Paragraph

4.2. Clauses not containing normative material sub-clause 2

5

EMISSION EVENT DATA MODEL

5 EMISSION EVENT DATA MODEL

The EmissionML conceptual model is described as a class diagram in Figure x. The schema is fully described in x.x.x.

Use the following table for Requirements Classes.

Table 2

Emission Event Conceptual Data Model Core	
http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core	
Target type	Data Model
Dependency	http://www.opengis.net/doc/as/om/3.0/req/obs-basic
Requirement 1	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/emission-event This requirement defines the Emission Event as a time-bounded occurrence of pollutant release, linking source, mechanism, quantity, and temporal bounds.
Requirement 2	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/source This requirement defines the Source as a persistent geospatial feature from which one or more emission events originate.
Requirement 3	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/mechanism This requirement defines the Mechanism as the causal process or pathway by which an emission occurs, independent of source identity.
Requirement 4	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/emission-quantity This requirement defines the Emission Quantity as the asserted amount of pollutant released, including unit, determination method, and supporting evidence.
Requirement 5	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/start-time This requirement defines the start time of an emission event as a determined temporal bound supported by a determination method and optional evidence.
Requirement 6	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/end-time This requirement defines the end time of an emission event as an optional determined temporal bound indicating cessation or reporting boundary.
Requirement 7	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/unit-of-measure This requirement defines the Unit of Measure used to express emission quantities through externally referenced, authoritative unit definitions.
Requirement 8	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/observation

	This requirement defines the Observation as evidence supporting emission quantity or temporal bound determinations, potentially referencing features of interest at varying spatial scales.
Requirement 9	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/temporal-bound This requirement defines the Temporal Bound as a determined temporal instant associated with an emission event, optionally supported by quality metadata and evidence.
Requirement 10	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/determination-method This requirement defines the Determination Method as the method used to derive emission quantities or temporal bounds.

5.1. Emission Event Class

An Emission Event represents the release of a substance into the atmosphere over a non-zero duration. It is characterized by its start time and end time, the Source Feature from which the substance originated, and the identity and quantity of the emitted substance.

Note: Conceptually, an Emission Event corresponds to an occurrent in upper-ontology terms, as it unfolds over time and has temporal parts. Note: For reporting or accounting purposes, a long-running emission event is often segmented into multiple time-bounded emission events (e.g., annual records). This segmentation reflects reporting conventions and does not imply physical cessation or re-initiation of the emission.

Table 3

Requirement 1: Emission Event Class	/req/datamodel/core/emission-event Each Emission Event class shall have the mandatory attributes listed in Table 4. Each Emission Event class shall have the direct relation between an Emission Event class and other classes listed in Table 5.

Table 4 — Attributes of an Emission Event class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
pollutant Type	The URI linking to an external definition of the pollutantType of the Emission Event.	URI	1 (mandatory)

Table 5 — Direct relation between an Emission Event class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
has Mechanism	The mechanism by which the emission occurred.	Mechanism	1 (mandatory)
hasSource	The source from which the emission originated.	Source	1 (mandatory)
startTime	The start time of the emission event.	TemporalBound	1 (mandatory)
endTime	The end time of the emission event.	TemporalBound	0..1 (optional)
hasQuantity	The quantity of the substance emitted.	EmissionQuantity	1 (mandatory)

5.2. Source Class

A Source represents the feature from which an emission originates. A Source is an ISO 191xx feature whose geometry provides the geospatial reference for one or more Emission Events. A Source is characterised by its own identity, geometry, intended function, and lifecycle, and is explicitly agnostic to the Mechanism by which any associated Emission Event occurs.

Note: A Source is a continuant that may exist independently of any emission, whereas an Emission Event is an occurrent. The identity and lifecycle of a Source are independent of the existence, absence, or frequency of Emission Events.

Table 6

Requirement 2: Source Class	/req/datamodel/core/source + Each Source class shall have the mandatory attributes listed in Table 7. Each Source class shall have the direct relation between a Source class and other classes listed in Table 8.

Table 7 — Attributes of a Source class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
name	A human-readable name for the source.	String	0..1 (optional)
geometry	The spatial geometry of the source.	GM_Object	1 (mandatory)

Table 8 — Direct relation between a Source class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
hasType	The type classification of the source.	SourceType	0..1 (optional)

5.3. Source Type Class

A Source Type provides a classification for Sources, describing the type or category of the Source.

Table 9

Requirement 3: Source Type Class	/req/datamodel/core/source-type + Each Source Type class shall have the mandatory attributes listed in Table 10.

Table 10 — Attributes of a Source Type class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
name	A human-readable name for the source type.	String	1 (mandatory)
description	A detailed description of the source type.	String	1 (mandatory)
definition	The URI linking to an external definition of the source type.	URI	1 (mandatory)

Table 11 — Direct relation between a Source Type class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
-	No direct relations are defined for SourceType in the core model (it is referenced by Source).	-	-

5.4. Mechanism Class

A Mechanism is a classification of the specific physical, chemical, or operational process that constitutes the direct cause or pathway for an emission. It is defined independently of the

Source—including its identity, structure, or purpose—allowing Mechanisms to be consistently reused across different types of Source.

Table 12

Requirement 4: Mechanism Class	/req/datamodel/core/mechanism + Each Mechanism class shall have the mandatory attributes listed in Table 13. Each Mechanism class shall have the direct relation listed in Table 14.

Table 13 — Attributes of a Mechanism class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
label	A human-readable label for the mechanism.	String	1 (mandatory)
description	A detailed description of the mechanism.	String	0..1 (optional)

Table 14 — Direct relation between a Mechanism class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
has Intent	The emission intent associated with the mechanism.	EmissionIntent	1 (mandatory)

5.5. Emission Quantity Class

An Emission Quantity represents the asserted quantity of pollutant released as part of an Emission Event.

Table 15

Requirement 5: Emission Quantity Class	/req/datamodel/core/emission-quantity + Each Emission Quantity class shall have the mandatory attributes listed in Table 16. Each Emission Quantity class shall have the direct relation between an Emission Quantity class and other classes listed in Table 17.

Table 16 — Attributes of an Emission Quantity class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
quantity	The numeric value of the emission quantity.	double	1 (mandatory)
quality	A data quality element describing the quality of the emission quantity value.	DQ_Element	0..* (optional)

Table 17 — Direct relation between an Emission Quantity class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
hasUnit	The unit of measure associated with the emission quantity.	UnitOfMeasure	1 (mandatory)
hasDetermination Method	The method used to determine or derive the emission quantity.	Determination Method	1 (mandatory)
hasEvidence	Observations that provide evidence supporting the emission quantity assertion.	Observation	0..* (optional)

5.6. Unit of Measure Class

A Unit of Measure is a reference quantity chosen from a unit equivalence group (Source: ISO/OGC 19156:2023)

Table 18

Requirement 6: Unit of Measure Class	/req/datamodel/core/unit-of-measure + Each Unit of Measure class shall have the mandatory attributes listed in Table 19.

Table 19 — Attributes of a Unit of Measure class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
name	A human-readable name for the unit of measure.	String	1 (mandatory)
symbol	A short symbol representing the unit of measure.	String	1 (mandatory)

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
definition	The URI linking to an external definition of the unit of measure.	URI	1 (mandatory)

Table 20 — Direct relation between a Unit of Measure class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
-	No direct relations are defined for UnitOfMeasure in the core model.	-	-

5.7. Temporal Bound Class

A Temporal Bound is a value object representing a temporal determination (e.g., start time or end time) with optional data quality metadata. Temporal Bounds may be supported by evidence Observations and are associated with a Determination Method.

Table 21

Requirement 7: Temporal Bound Class	/req/datamodel/core/temporal-bound + Each Temporal Bound class shall have the mandatory attributes listed in Table 22. Each Temporal Bound class shall have the direct relation between a Temporal Bound class and other classes listed in Table 23.

Table 22 — Attributes of a Temporal Bound class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
time	The time instant value of this Temporal Bound.	TM_Instant	1 (mandatory)
quality	A data quality element describing the quality of the temporal bound.	DQ_Element	0..* (optional)

Table 23 — Direct relation between a Temporal Bound class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
hasDetermination Method	The method used to determine or derive the temporal bound.	Determination Method	1 (mandatory)

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
hasEvidence	Observations that provide evidence supporting the temporal bound assertion.	Observation	0..* (optional)

5.8. Observation Class

In EmissionML, an Observation is evidence that may support an Emission Quantity or Temporal Bound determination.

Note: The Ultimate Feature of Interest of an Observation assists in identifying the Source of the associated Emission Event, without requiring the Feature of Interest to coincide exactly with the Source. It is common for a single Emission Event to be observed by multiple Observations whose Ultimate Features of Interest are defined at different spatial scales. For example, a satellite remote sensing image Observation typically has an area-based Ultimate Feature of Interest (e.g., a site), whereas the same Emission Event may also be observed by a person carrying an Optical Gas Imaging camera, for which the Ultimate Feature of Interest is a specific component of an item of equipment from which pollutants are released.

Table 24

Requirement 8: Observation Class	/req/datamodel/core/observation + Each Observation class shall be a ISO/OGC 19156:2023 Observation.

5.9. Emission Intent Enumeration

Emission Intent is an enumeration describing whether an emission is intentional, unintentional, or unknown.

Table 25

Requirement 9: Emission Intent Enumeration	/req/datamodel/core/emission-intent + The Emission Intent enumeration shall support the enumeration values listed in Table 26.

Table 26 — Enumeration values of Emission Intent

NAME	DEFINITION
INTENTIONAL	The emission occurred intentionally.
UNINTENTIONAL	The emission occurred unintentionally.
UNKNOWN	The intent of the emission is unknown.

5.10. Determination Method Class

A Determination Method describes how a value (e.g., an emission quantity or temporal bound) was determined.

Table 27

Requirement 10: Determination Method Class	/req/datamodel/core/determination-method + Each Determination Method class shall have the mandatory attributes listed in Table 28.

Table 28 — Attributes of a Determination Method class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
label	A human-readable label for the determination method.	String	1 (mandatory)
description	A detailed description of the determination method.	String	0..1 (optional)
definition	The URI linking to an external definition of the determination method.	URI	0..1 (optional)

A

ANNEX A (INFORMATIVE) BIBLIOGRAPHY

A ANNEX A (INFORMATIVE) BIBLIOGRAPHY

NOTE: The TC has approved Springer LNCS as the official document citation type.

Springer LNCS is widely used in technical and computer science journals and other publications

– Actual References:

[n] Journal: Author Surname, A.: Title. Publication Title. Volume number, Issue number, Pages Used (Year Published)

[1] OGC: OGC Testbed 12 Annex B: Architecture. (2015).